

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage:20%

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

3. Motion Analysis Decision Problem

A traffic system requires:

- Accurate motion direction detection
- Robustness in dense traffic

Results:

Optical Flow: accurate but noisy in dense scenes

Motion Estimation: stable but less detailed

Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.

MOTION ANALYSIS DECISION PROBLEM

Comparison of Optical Flow and Motion Estimation

Criterion	Optical Flow	Motion Estimation
Motion direction detection	Highly accurate	Accurate but less detailed

Performance in dense traffic	Can become noisy due to overlapping objects	More stable in crowded scenes
Motion detail	Provides detailed pixel-level motion information	Provides overall/object-level motion information
Noise sensitivity	High	Lower
Computational complexity	Relatively high	Generally lower
Suitability for crowded traffic	Moderate	High

Scenario-Based Evaluation

Scenario 1 – Low traffic density:

When only a few vehicles are present, **Optical Flow** is preferable because individual vehicle movements can be tracked precisely. It provides detailed information about the direction and magnitude of motion.

Scenario 2 – High traffic density:

When many vehicles are close together, the motion patterns of different vehicles overlap. Optical Flow may produce noisy or ambiguous vectors, making accurate direction detection difficult. **Motion Estimation** is more stable because it focuses on overall motion patterns and is less affected by pixel-level interference.

Scenario 3 – Traffic monitoring at a crowded intersection:

At a busy intersection, vehicles may partially occlude one another and move in different directions. In this situation, the stability of Motion Estimation is more valuable than the additional detail provided by Optical Flow.

Decision and Justification

For the given requirement of **accurate motion direction detection under high-density traffic conditions**, **Motion Estimation is the more suitable approach**. Although Optical Flow provides more detailed motion information and can be highly accurate in simple scenes, its sensitivity to overlapping movements makes it noisy in dense traffic.

Therefore, **Motion Estimation should be selected as the primary technique for high-density traffic monitoring**, because its greater stability and resistance to motion interference provide more reliable direction analysis in crowded scenarios. Optical Flow can still be used when detailed motion information is required in less congested environments.

Final Decision: → **Motion Estimation**

Reason: *In high-density traffic, stability and robustness are more important than pixel-level motion detail*

9. Mobile Face Detection Trade-off

Constraints:

- Real-time operation required
- Accuracy $\geq 85\%$

Options:

Viola-Jones: 80%, fast

Deep Learning: 95%, slow

Compare and evaluate which approach should be chosen. Justify using constraint satisfaction logic.

MOBILE FACE DETECTION TRADE-OFF

Comparison

Criterion	Viola-Jones	Deep Learning
Accuracy	80%	95%
Speed	Fast	Slow
Real-time operation	Satisfied	May not be satisfied
Accuracy $\geq 85\%$	✗ Not satisfied	✓ Satisfied
Overall constraint satisfaction	✗	✓

Constraint Satisfaction Logic

There are **two mandatory constraints**:

1. **Real-time operation is required.**
2. **Accuracy must be at least 85%.**

Viola-Jones is fast and therefore satisfies the real-time requirement. However, its accuracy is only **80%**, which is below the required **85%**. Therefore, it **fails one mandatory constraint** and cannot be selected.

Deep Learning provides **95% accuracy**, which exceeds the required 85%. Although it is slower than Viola-Jones, it is the only option that satisfies the critical accuracy constraint. If the system is optimized using mobile hardware acceleration, model compression, or lightweight architectures, its processing speed can be improved to support real-time operation.

Decision

Deep Learning should be chosen, because **accuracy $\geq 85\%$ is a hard constraint**, and Viola-Jones fails it.

Final Decision: → **Deep Learning**

Justification: Viola-Jones is faster but does not meet the minimum accuracy requirement ($80\% < 85\%$). Deep Learning achieves 95% accuracy and therefore satisfies the accuracy constraint; its speed can be optimized for real-time mobile deployment.

18. Crowd Monitoring System

Requirements:

- Stable detection in dense crowds
- Moderate accuracy acceptable

Options:

Optical Flow: unstable in dense scenes Motion Models: stable

Compare and evaluate. Justify your decision.

CROWD MONITORING SYSTEM

Comparison

Criterion	Optical Flow	MotionModels
Stability in dense crowds	✗ Unstable	✓ Stable
Accuracy	High detail but affected by crowd overlap	Moderate
Handling occlusion	Less reliable	More reliable
Suitability for dense crowds	Low	High

Requirement satisfaction	✗	✓
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Evaluation

Optical Flow provides detailed information about the movement of individual pixels. However, in dense crowds, people frequently overlap and move in different directions. This can produce noisy or inconsistent motion vectors, making detection unstable.

Motion Models focus on the overall movement patterns of objects or groups. They are more stable in crowded environments because they are less affected by individual pixel-level changes and partial occlusions.

Scenario-Based Justification

Consider a **crowded railway station or stadium exit** where hundreds of people are moving simultaneously. Optical Flow may generate overlapping and noisy motion vectors, resulting in unstable detection. A Motion Model can identify the general movement pattern of the crowd more consistently.

Since the requirement specifies that **stable detection is more important**, and **moderate accuracy is acceptable**, the stability advantage of Motion Models outweighs the additional detail provided by Optical Flow.

Final Decision

→ **Motion Models should be selected.**

Justification: *Motion Models are more suitable for dense crowd monitoring because they provide stable detection despite overlapping movements and occlusions. Since only moderate accuracy is required, their stability makes them the better choice over Optical Flow.*

20. Smart Parking System

Requirements:

- Detect vehicles accurately
- Low computational cost

Options:

Viola-Jones: low cost, low accuracy

YOLO: moderate cost, high accuracy

Compare and evaluate. Justify your decision.

SMART PARKING SYSTEM

Comparison

Criterion	Viola-Jones	YOLO
Vehicle detection accuracy	Low	High
Computational cost	Low	Moderate
Detection reliability	Low	High
Suitability for parking system	Moderate	High
Accuracy requirement	 Not sufficiently satisfied	 Satisfied
Cost requirement	 Satisfied	 Moderate

Evaluation

Viola-Jones has low computational cost, making it suitable for systems with limited processing power. However, its low accuracy can cause vehicles to be missed or incorrectly detected. In a smart parking system, this may result in incorrect information about available or occupied parking spaces.

YOLO has a moderate computational cost but provides **high vehicle-detection accuracy**. It can detect multiple vehicles efficiently and is more reliable in practical parking environments where vehicles may have different sizes, orientations, and positions.

Scenario-Based Justification

Consider a **busy parking area** where several vehicles enter and leave simultaneously. If Viola-Jones incorrectly detects a vehicle, the system may display an occupied slot as available or vice versa. This directly affects the reliability of the parking system.

YOLO can detect multiple vehicles with higher accuracy, making the parking-slot status more dependable. Although its computational cost is higher, it remains **moderate** rather than excessive, making the trade-off acceptable when accurate detection is important.

Final Decision

→ **YOLO should be selected.**

Justification: *Although Viola-Jones has lower computational cost, its low accuracy can compromise vehicle detection. YOLO provides substantially higher accuracy with only moderate computational cost, making it the better overall choice for a smart parking system where reliable vehicle detection is essential.*

21. Medical Imaging Analysis

Requirements:

- Very high accuracy (>95%)
- Processing time not critical

Options:

Traditional: 85%

Deep Learning: 97%

Compare and evaluate the suitable approach. Justify your decision.

MEDICAL IMAGING ANALYSIS

Comparison

Criterion	Traditional Methods	Deep Learning
Accuracy	85%	97%
Required accuracy >95%	✗ Not satisfied	✓ Satisfied
Processing time	Faster	Slower
Importance of processing time	Not critical	Not critical
Overall suitability	Low	High

Evaluation

Traditional image-processing methods achieve only **85% accuracy**, which is significantly below the required accuracy of **more than 95%**. Therefore, they fail the most important constraint.

Deep Learning achieves **97% accuracy**, exceeding the required threshold. Although it may require more processing time and computational resources, processing time is explicitly **not critical** for this application. Therefore, the slower processing does not prevent its selection.

Scenario-Based Justification

Consider a medical imaging system used to identify abnormalities in **X-rays, CT scans, or MRI images**. High accuracy is essential because incorrect classification can lead to missed abnormalities or incorrect diagnoses.

In this scenario, a method with 85% accuracy is unsuitable even if it processes images faster. Deep Learning, with **97% accuracy**, provides the required reliability, while its longer processing time is acceptable because immediate results are not a requirement.

Final Decision

→ **Deep Learning should be selected.**

Justification: *Deep Learning satisfies the critical accuracy requirement with 97% accuracy, while its slower processing is acceptable because processing time is not critical. Traditional methods fail the required >95% accuracy constraint.*

27. Robotics Vision System

Requirements:

- Fast decision-making
- Moderate accuracy

Options:

Deep Learning: slow

Traditional: fast

Compare and evaluate. Justify your decision.

ROBOTICS VISION SYSTEM

Comparison

Criterion	Deep Learning	Traditional Methods
Decision-making speed	Slow	Fast
Accuracy	High	Moderate
Required accuracy	Moderate	Satisfied
Real-time response	✗ Less suitable	✓ Suitable
Overall suitability	Low	High

Evaluation

Deep Learning can provide high accuracy and robust visual recognition, but its slower processing speed can delay decisions. In robotics, delayed decisions may affect the robot's ability to respond to obstacles, objects, or changes in its environment.

Traditional vision techniques are generally faster and require fewer computational resources. Since the requirement specifies only **moderate accuracy**, their accuracy is sufficient for the application.

Scenario-Based Justification

Consider a **robot navigating a warehouse**. The robot must quickly detect obstacles and decide whether to move, stop, or change direction. A slow vision system could cause delayed responses and increase the risk of collision.

Traditional methods can process visual information quickly and provide sufficiently accurate results for basic object or obstacle detection. Therefore, their speed is more valuable than the additional accuracy offered by Deep Learning.

Final Decision

→ **Traditional Vision Methods should be selected.**

Justification: *Fast decision-making is the critical requirement, while only moderate accuracy is required. Traditional methods satisfy both conditions more effectively, whereas Deep Learning's slower processing makes it less suitable for time-sensitive robotic applications.*

30. Motion Animation System

Requirements:

- Smooth motion
- Real-time rendering

Options:

Motion Models: realistic, slow

Motion Estimation: fast, less realistic

Compare and evaluate design approach. Justify your decision.

MOTION ANIMATION SYSTEM

Comparison

Criterion	Motion Models	Motion Estimation
Motion smoothness / realism	High	Moderate
Rendering speed	Slow	Fast
Real-time rendering	✗ Difficult	✓ Suitable
Visual realism	High	Lower
Overall suitability	Moderate	High

Evaluation

Motion Models can produce realistic and smooth motion, making them attractive when visual quality is the primary requirement. However, their slower processing can cause delays and reduce the frame rate during real-time rendering.

Motion Estimation is computationally faster and can process motion efficiently enough for real-time applications. Although the resulting motion may be less realistic, it can still provide sufficiently smooth animation when optimized for high frame rates.

Scenario-Based Justification

Consider a **real-time 3D animation or interactive simulation** where the system must continuously render moving objects. If Motion Models are used, their slow processing may introduce lag, frame drops, or noticeable delays. This directly violates the real-time rendering requirement.

Motion Estimation can generate motion information quickly, allowing the system to maintain responsive and continuous rendering. The reduced realism is an acceptable trade-off when real-time performance is the priority.

Final Decision

→ **Motion Estimation should be selected.**

Justification: *Real-time rendering is the critical requirement. Although Motion Models provide more realistic motion, their slow processing can cause rendering delays. Motion Estimation provides faster processing and better real-time responsiveness, making it the more suitable design approach.*

Code of Conduct:

I P.V.N. ANISH BABU (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.V.N. ANISH